

A Machine-Checked Analysis of a Cone-Algebra Model for Neural Population Computation

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IPNPCNS formalization project

A technical account for readers with mathematical maturity but no assumed knowledge of Lean.

Abstract

The model under study represents ensembles of neural population messages by closed convex cones in a real Hilbert space. Population transformations are then described using cone sum, intersection, reflection, metric projection, and a residual operation called conic rejection. This report gives a self-contained account of a Lean formalization of that mathematical program.

The central exact results are machine-checked for arbitrary closed convex cones in arbitrary real Hilbert spaces. They include Moreau decomposition, the identity that conic projection is two successive conic rejections, and three rejection-only formulas for cone intersection. The development also verifies the corresponding finite-subspace and Moore–Penrose formulas, deterministic near-isometry and sparse Gram bounds, finite nonnegative least-squares optimality and convergence, a finite collateral-occupancy model, an exact orthonormal wavelet example, a compact causal Volterra filter, a quantified membrane-update remainder, and conditional population-level realization theorems.

A second purpose is methodological. The report follows one central identity from paper notation to an exact Lean declaration and explains enough of the language and trusted checker for a non-Lean reader to inspect the claim. It also distinguishes source mathematics from explicit formalization choices, new supporting lemmas, constructive strengthenings, and concrete models added by the Lean development.

The verification has a precise boundary. It proves mathematical consequences of explicit assumptions; it does not prove that biological neural populations satisfy those assumptions. Stochastic learning, generic Laplace-domain filter realization, random embedding probabilities, support selection by biological dynamics, and recurrent-network stability remain outside the compiled theorem set. An equation-by-equation audit against the LaTeX source found no material discrepancy in the claims made by the Lean project. A full build contains no admitted proofs or custom axioms, and an axiom audit across the principal endpoints reports only the standard foundational principles used by Lean and mathlib.

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1 Question, answer, and scope

The mathematical proposal examined here begins with a change of object. A single activity vector records one message from one neural population. An ensemble of related messages, however, is represented by the closed convex cone generated by those vectors. The invariant is therefore not a point but a geometric set. A population may transform the coordinates of individual messages while retaining the algebraic relations among the cones that those messages generate.

This proposal creates three logically different questions:

1. Are the definitions coherent, and do the claimed mathematical identities follow from the stated hypotheses?
2. Do the idealized population and learning interfaces imply the claimed cone operations?
3. Do real neural systems satisfy those interfaces with biologically meaningful error levels?

Lean can settle the first question and conditional versions of the second. The third is empirical. No proof assistant can infer anatomical connectivity, time-scale separation, or in-vivo convergence from pure mathematics.

Overall conclusion

The deterministic mathematical core is realistic to formalize and has been formally verified at substantial breadth. The headline cone identities hold without finite-dimensionality or finite-generation assumptions. The population interpretation is verified only as an implication from explicit model premises, not as a theorem about the central nervous system.

The report is organized by mathematical dependence rather than by the order in which files were written. Subspaces provide a linear benchmark. Closed cones then supply the central geometry. Approximate invariance, temporal signals, learning, occupancy, and population realization form separate layers around that core. The final sections explain exactly what the machine check certifies and where it stops.

1.1 Four kinds of formal status

Throughout the report, claims are classified as follows.

Table 1: Interpretation of verification status.

Status	Meaning
Verified	A Lean theorem derives the stated conclusion from assumptions no stronger than those shown. A formally stronger theorem may cover the paper statement as a specialization.
Conditional	The mathematical implication is proved, but a regularity, sampling, training, or biological premise is made explicit rather than derived.
Represented	An object or relation is encoded as an interface so that later consequences can be checked, but the interface itself is not established by the development.
Outside scope	The item is empirical, cited background, underspecified stochastic theory, or a deliberately excluded mathematical extension. No verification claim is made for it.

Verification status and source provenance answer different questions. A theorem may be fully verified while still being a new strengthening supplied by the formalization; conversely, a formula may be

copied directly from the paper but remain outside the verified scope. The following provenance vocabulary is therefore used independently of table 1.

Table 2: Provenance of mathematics in the formal development.

Provenance	Meaning
Paper claim	A definition, equation, or theorem stated in the paper and represented without material change.
Explicit interpretation	A domain, closure, sign, boundary, or regularity condition left implicit in prose and made visible in the formal statement.
Formalization lemma	New mathematical infrastructure or decomposition used to reach a source claim, but not attributed to the paper.
Strengthening	A new result that implies, quantifies, or extends the source claim; its additional content belongs to the formalization.
Concrete realization	A particular finite or analytic model that satisfies an informal interface. It demonstrates consistency and consequences, not empirical truth or uniqueness of interpretation.

Table 3 gives the high-level division. This division is not cosmetic: it prevents a machine-checked theorem newly discovered during formalization from being retrospectively presented as a theorem of the paper.

Table 3: Where the principal mathematical content originates.

Layer	From the paper	Added or made explicit in Lean
Closed-cone algebra	Closed conic hull, polar, projection, rejection, double rejection, and intersection formulas	A closure-safe representation, the complete metric-projection and Moreau lemma stack, and named pointwise-to-hull lifting lemmas
Finite subspaces	Matrix and Moore–Penrose renderings of projection, rejection, and comparison	A general rectangular pseudoinverse construction from the effective map, including all four Penrose equations and uniqueness
Learning	NNLS objective, projected update, and informal convergence under “standard assumptions”	Exact KKT equivalence and separate full-rank and rank-deficient deterministic convergence theorems; stochastic convergence is not supplied
Signals	Finite-window signal and filter equations, wavelet example, and low-pass approximation	A selected L^2 boundary convention, compact causal Volterra model, exact orthonormality proofs, and a quantified quadratic membrane remainder
Occupancy and populations	Occupancy approximation, overlap heuristic, active faces, and circuit equations	An exact finite sampling experiment, uniform exponential error bound, independent product construction, and sufficient active-face certificates

2 From population messages to Hilbert-space geometry

2.1 Messages and invariants

Let H be a real Hilbert space. In the static finite-dimensional model, $H = \mathbb{R}^n$ and a message is a vector of population activities. In the temporal model,

$$H = L^2([0, T]; \mathbb{R}^n),$$

so each neuronal coordinate is itself a time-varying square-integrable signal. A finite family of messages $v_1, \dots, v_m$ generates the cone

$$C = \text{Conic}\{v_1, \dots, v_m\} = \left\{ \sum_{i=1}^m \lambda_i v_i : \lambda_i \geq 0 \right\}.$$

For an arbitrary set $S \subseteq H$, the source uses the more general definition

$$\text{Conic}(S) = \overline{\left\{ \sum_{i=1}^k \lambda_i x_i : k \in \mathbb{N}, x_i \in S, \lambda_i \geq 0 \right\}},$$

where closure is in the norm topology. The closure is mathematically important: pointwise images and Minkowski sums of closed cones need not be closed in unrestricted infinite-dimensional settings.

The formal development therefore separates two levels.

- Neural representations are usually finitely generated. Their spans remain finite-dimensional even if the ambient signal space is not.
- The main cone identities are stated for arbitrary closed convex cones in arbitrary real Hilbert spaces.

This separation both matches the source and strengthens the central results: the proofs do not depend on a chosen finite frame.

2.2 The operations

For a closed cone C , write $P_C x$ for the unique metric projection of x onto C . We use C° for the source's non-positive polar:

$$C^\circ = \{z \in H : \langle z, y \rangle \leq 0 \text{ for every } y \in C\}.$$

The source prints this object as $\neg C$. The present report uses C° to avoid confusing unary polarity with the binary rejection operation.

The operation $A \sqcup B$ is not merely the pointwise image $\{P_B a : a \in A\}$. The metric projection onto a cone is generally nonlinear, and its image of another cone need not be convex. Taking the closed conic hull is part of the definition.

2.3 Where verification sits

Figure 1 summarizes the logical boundary. Arrows mean *is used as a premise for*, not *is proved by*. The formal checker reaches the top two mathematical layers. The empirical layer remains a scientific validation problem.

Table 4: Closed-cone operations used by the model. The report writes $A \searrow B$ for the paper’s binary conic-rejection notation.

Operation	Notation	Definition
Polar	C°	All vectors pairing non-positively with every point of C .
Closed sum	$A + B$	$\text{Conic}\{a + b : a \in A, b \in B\}$.
Reflection	$-C$	$\{-c : c \in C\}$.
Conic projection	$A \lrcorner B$	$\text{Conic}\{P_B a : a \in A\}$.
Conic rejection	$A \searrow B$	$\text{Conic}\{a - P_B a : a \in A\} = A \lrcorner B^\circ$.
Intersection	$A \cap B$	Ordinary set intersection, again a closed convex cone.

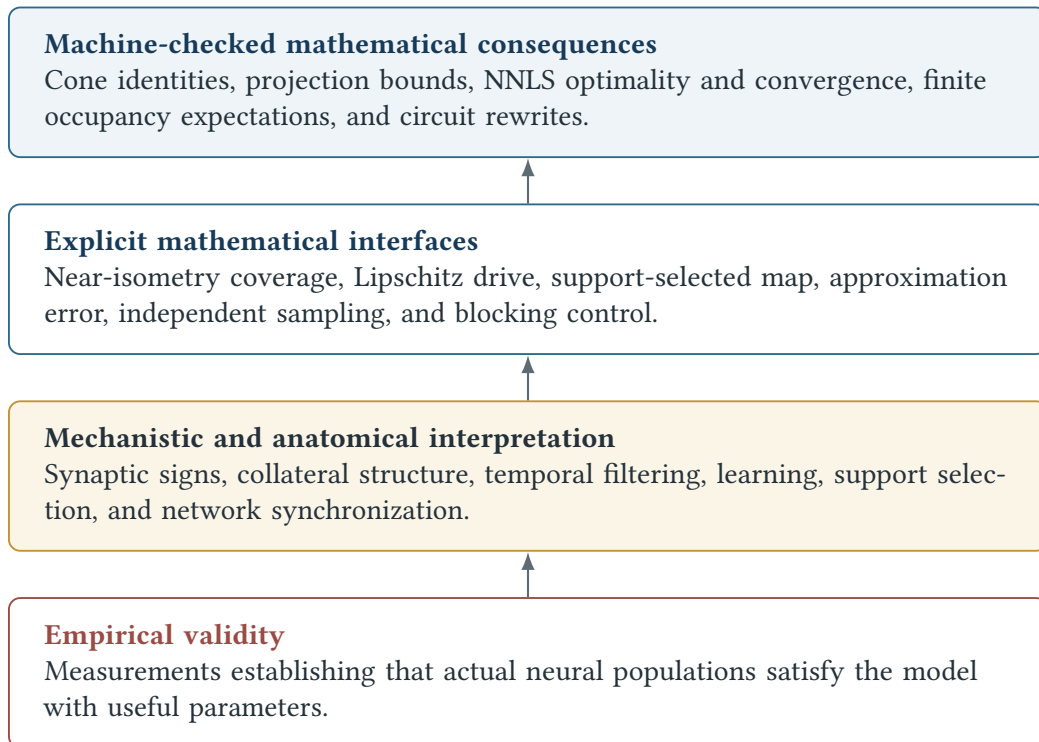


Figure 1: Verification boundary. Lean checks consequences above the empirical layer; it does not convert model interpretation into biological evidence.

3 What the machine check means

3.1 From paper prose to a checked declaration

Lean is an interactive theorem prover based on dependent type theory [7]. The useful operational picture is simpler than that foundation may suggest. A *definition* assigns an exact meaning to a name. A *theorem declaration* states a proposition as a type. Its proof is a term of that type. Commands called *tactics* help construct the term, an *elaborator* resolves notation and omitted arguments, and a small *kernel* independently checks the fully explicit result. The mathematical library *mathlib* contributes previously checked definitions and theorems for Hilbert spaces, convexity, measure theory, matrices, topology, and finite combinatorics [8].

The paper’s double-rejection identity provides a running example. In the notation of this report it says

$$A \perp B = A \setminus_{\perp} (A \setminus_{\perp} B).$$

Before this can be a Lean proposition, every symbol must have a fixed type and meaning. The project first declares the two operations:

Listing 1: Definitions behind the running example, with a helper name inlined.

```
def coneProjection (A B : ClosedCone H) : ClosedCone H :=
  conicHull (metricProjection B '' (A : Set H))

def coneRejection (A B : ClosedCone H) : ClosedCone H :=
  coneProjection A (polar B)
```

Thus projection of one cone onto another is not assumed to be a cone: the pointwise image is deliberately wrapped in the norm-closed conic hull. Rejection is projection onto the paper’s non-positive polar. These choices are the formal contract for source equations (73)–(75), (98), and (108).

The public theorem then reads:

Listing 2: The public Lean endpoint for double rejection.

```
theorem doubleRejection (A B : ClosedCone H) :
  coneProjection A B =
    coneRejection A (coneRejection A B) := by
  -- prove both inclusions between the two closed cones
  apply le_antisymm
  ...
```

The parameters A and B are closed convex cones in a previously declared space H . The type-class assumptions on H , printed elsewhere as bracketed arguments, say that it is a complete real inner-product space. The first colon separates the theorem’s parameters from its proposition; `:= by` begins the proof. Equality here is extensional: the two closed cones have exactly the same members. The line `apply le_antisymm` reduces equality of ordered cone objects to the two inclusions. Later `rw` steps replace a metric projection using a proved pointwise lemma. The ellipsis abbreviates this report’s listing, not the checked file: the source contains the complete proof and no placeholder.

3.2 What is translated, and what is designed

The route from the printed equation to the theorem is not mechanical textual translation. It has five inspectable stages:

1. **Source reading.** The PDF and LaTeX source identify the intended operations, the non-positive polar convention, and the generality of closed cones in a Hilbert space.

2. **Mathematical restatement.** The equation is separated from its biological interpretation and written as equality of closed cones.
3. **Representation design.** `ClosedCone`, `H`, `conicHull`, `polar`, `coneProjection`, and `coneRejection` decide how the mathematics inhabits Lean and mathlib. Closure and completeness become explicit.
4. **Proof architecture.** New formalization lemmas establish metric projection and Moreau decomposition, prove the pointwise projection identity, and lift equality of generator sets through the same closed-hull constructor.
5. **Kernel certification.** Lean elaborates the declaration and proof to a fully explicit term. The kernel checks that term against the imported definitions and theorem types.

Stages 1 and 2 aim to preserve paper mathematics. Stage 3 necessarily contains modeling and library-integration choices. Stage 4 contains genuine mathematics added by the formalization, even when its purpose is to prove a paper theorem. Stage 5 verifies derivability of the resulting contract. The independent source audit closes the loop from stage 5 back to stage 1.

The guarantee for the running example

The kernel check establishes that, under the displayed Hilbert-space and closed-cone assumptions and the project's exact definitions, double rejection equals conic projection. The source audit establishes that this contract matches the relevant paper equations. Neither check establishes that a neural circuit implements the abstract operations or that the model is biologically adequate.

3.3 How library facts and local proofs compose

Imports are not appeals to an unchecked handbook. Lean records the types of the mathlib theorems used by a local proof, and the kernel checks the final term through that dependency graph. Some low-level implementation details remain in Lean's trusted computing base—the kernel, elaborated environment, compiler and runtime used to run the checker—but tactics themselves need not be trusted: a faulty tactic can fail or construct a rejected term, not make an invalid term pass the kernel.

This distinction also explains the value of named helper lemmas. A declaration such as `metricProjection_polar_rejection_eq` records the central pointwise step as a reusable mathematical interface. The public theorem can then be read as a short composition of checked facts rather than as a transcript of tactic search.

3.4 Classical and noncomputable mathematics

Metric projection in an arbitrary Hilbert space is selected from an existence theorem. The formal definition is therefore marked noncomputable. This is standard classical mathematics: it permits reasoning about the unique minimizer without claiming executable code for finding it. Uniqueness later makes all observable results independent of the particular witness chosen.

The axiom audit reports only:

Propositional extensionality.

Logically equivalent propositions may be identified.

Classical choice.

A witness may be selected from a proof that a suitable object exists.

Quotient soundness.

Equivalent representatives define the same quotient object.

These are standard foundations of Lean and mathlib. The development adds no custom axiom and contains no admitted proof placeholder.

3.5 What compilation does not mean

A successful build establishes type correctness and proof checking for the formal statements actually present. It does not establish that those statements are faithful to the source merely because their names resemble equation numbers. Faithfulness was addressed separately by reconstructing the formulas and qualifying prose from the LaTeX source. It also does not establish numerical stability of an implementation, because the central metric projection is a mathematical object rather than extracted software.

4 The linear benchmark: finite subspaces

The paper introduces subspaces before cones. This layer is not dispensable background: it fixes matrix orientation, projection signs, rejection conventions, and comparison measures in a setting where familiar linear algebra provides strong regression tests.

Provenance. The matrix and subspace identities in this section are source claims. Their coordinate-free organization is a formalization choice; the rank-unrestricted Moore–Penrose construction and uniqueness proof are new supporting mathematics needed because the pinned library supplied no general matrix API.

4.1 Column spaces and the covariance rendering

For a finite real matrix A , the represented subspace is $\text{Col}(A)$. The source derives

$$\text{Col}(A) = \text{Col}(AA^\top).$$

The Lean proof is coordinate-free at its core: the range of a linear map equals the range of its composition with its adjoint. The finite matrix statement follows by translating matrices to Euclidean linear maps. It holds for rectangular, zero, and rank-deficient matrices.

Subspace sum is represented by concatenating generators. Orthogonal complement, projection, and rejection are represented by projector composition. A two-dimensional exact counterexample confirms that sum does not distribute over intersection.

4.2 A Moore–Penrose inverse without a library black box

The required version of mathlib did not provide a general bundled Moore–Penrose matrix API. The formalization therefore constructs the pseudoinverse mathematically.

For a finite-dimensional linear map $A : E \rightarrow F$, restrict A to $(\ker A)^\perp$. This restriction is a linear equivalence

$$A_{\text{eff}} : (\ker A)^\perp \simeq \text{range}(A).$$

The inverse map is composed with orthogonal projection onto $\text{range}(A)$, giving $A^+ : F \rightarrow E$. The development proves all four Penrose equations:

$$\begin{aligned} AA^+A &= A, & A^+AA^+ &= A^+, \\ (AA^+)^* &= AA^+, & (A^+A)^* &= A^+A, \end{aligned}$$

and proves uniqueness from those equations. Consequently, AA^+ is the orthogonal projector onto the column space and A^+A is the projector onto the row space.

This construction closes a common informal gap. Formulas containing A^+ are not restricted to full-rank matrices unless that restriction is explicitly stated.

4.3 Subspace operations and comparison

Let P, Q be closed subspaces. Intrinsic definitions use orthogonal projectors rather than matrices:

$$\begin{aligned} P \perp Q &= \text{range}(\pi_Q \circ \pi_P), \\ P \setminus_\perp Q &= P \perp Q^\perp. \end{aligned}$$

The verified identities include

$$P \perp Q = P \setminus_\perp (P \setminus_\perp Q)$$

and

$$P \cap Q = (P + Q) \setminus_\perp ((Q \setminus_\perp P) + (P \setminus_\perp Q)).$$

Matrix corollaries recover the source's pseudoinverse renderings.

For comparison, the development proves that the Frobenius pairing of projector matrices equals a basis-independent overlap:

$$\langle \Pi_P, \Pi_Q \rangle_F = \text{tr}(\Pi_P \Pi_Q) = \sum_{e \in \mathcal{B}_Q} \|\pi_P e\|^2,$$

for any orthonormal basis $\mathcal{B}_Q$ of Q . Normalizing this quantity yields the directional and symmetric subspace measures in the source.

5 Closed cones, metric projection, and Moreau decomposition

Provenance. The paper states the closed-cone and Moreau ingredients used by its algebra, with Moreau's theorem as classical background. The Lean project adds a complete local bridge from Hilbert projection to the exact sign and closure conventions required by the source; these bridge lemmas are not claimed as new content of the paper.

5.1 Metric projection

Let C be a nonempty closed convex cone in a real Hilbert space H . The Hilbert projection theorem gives a unique point

$$P_C x = \arg \min_{y \in C} \|x - y\|.$$

The formalization first selects a minimizer and proves membership and minimality. It then proves the variational characterization:

$$y = P_C x \iff y \in C \text{ and } \langle x - y, w - y \rangle \leq 0 \quad (\forall w \in C).$$

Because C is a cone, scaling test points turns this into the sharper cone-polar characterization

$$y = P_C x \iff y \in C, \quad x - y \in C^\circ, \quad \langle y, x - y \rangle = 0.$$

5.2 Moreau decomposition

The central decomposition is Moreau's classical cone theorem [2]; the formal result below includes the uniqueness form used throughout the later algebra.

Theorem 5.1 (Moreau decomposition). *For every closed convex cone $C \subseteq H$ and every $x \in H$,*

$$x = P_C x + P_{C^\circ} x, \quad \langle P_C x, P_{C^\circ} x \rangle = 0.$$

Conversely, every decomposition $x = y + z$ with $y \in C$, $z \in C^\circ$, and $\langle y, z \rangle = 0$ has $y = P_C x$ and $z = P_{C^\circ} x$.

The proof combines the projection characterization with bipolarity $(C^\circ)^\circ = C$. A particularly useful corollary is

$$P_{C^\circ}x = x - P_Cx.$$

This residual identity connects geometric projection to the model's rejection operation.

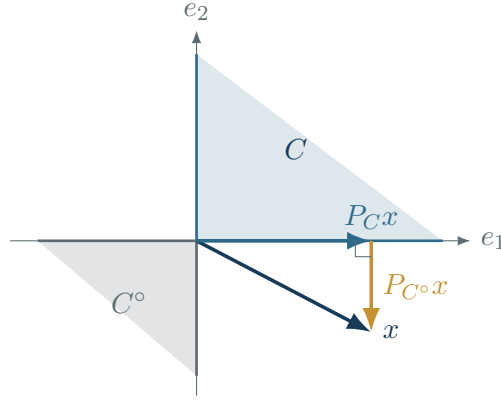


Figure 2: A two-dimensional Moreau decomposition. The cone component and polar component are orthogonal and add to the original vector. The theorem holds in arbitrary real Hilbert spaces.

5.3 Structural projection laws

Two further source claims are verified globally:

$$P_C(\alpha x) = \alpha P_Cx \quad (\alpha \geq 0)$$

and

$$P_Cx = P_C(P_{\text{span}(C)}x).$$

The second identity says that components orthogonal to the real span of the cone do not affect projection. For a finitely generated cone, the development also proves that this span equals the span of its generators and is finite-dimensional even when H is infinite-dimensional.

6 The two headline cone identities

Provenance. The double-rejection and intersection identities are the paper's headline mathematical claims. The pointwise projection lemma, closed-hull lifting argument, and decomposition into reusable order lemmas are formalization contributions that certify those claims without changing their generality.

6.1 Projection by two rejections

Theorem 6.1 (Double rejection). *For arbitrary closed convex cones A, B in a real Hilbert space,*

$$A \perp B = A \setminus_{\perp} (A \setminus_{\perp} B).$$

The pointwise argument reveals why the identity is true. Fix $x \in A$ and write its Moreau decomposition relative to B :

$$x = y + z, \quad y = P_Bx, \quad z = P_{B^\circ}x.$$

Let $R = A \setminus_{\perp} B$. By definition, $z \in R$, and $R \subseteq B^\circ$. Polarity reverses inclusion, so

$$B = (B^\circ)^\circ \subseteq R^\circ.$$

Thus $y \in R^\circ$, $z \in R$, and $y \perp z$. Moreau uniqueness, now applied to the cone R° , gives

$$P_B x = y = P_{R^\circ} x.$$

This equality holds for every generator $x \in A$. Taking the closed conic hulls of the identical pointwise generator sets yields the theorem.

Why closure handling matters

The proof does not claim that a pointwise projection image is already a closed convex cone. It first proves equality of the pointwise generators and then lifts that equality through the same closed-conic-hull constructor on both sides.

6.2 Intersection by rejections and sums

Define

$$R = (B \setminus_\perp A) + (A \setminus_\perp B), \quad K = R^\circ.$$

The source's three formulas are:

$$\begin{aligned} A \cap B &= A \setminus_\perp R, \\ &= B \setminus_\perp R, \\ &= (A + B) \setminus_\perp R. \end{aligned}$$

Theorem 6.2 (Rejection-only intersection). *All three formulas above hold for arbitrary closed convex cones in an arbitrary real Hilbert space.*

The longer proof has two geometric stages.

1. Since every common point of A and B is polar to both directed rejection cones, $A \cap B \subseteq K$. For $q \in K$, four Moreau decompositions relative to A , B , and their directed residuals imply

$$P_A q = P_B q = P_{A \cap B} q.$$

The central estimate combines polar signs with a completed-square identity; nonpositivity and nonnegativity force both directed residuals to vanish.

2. Project a point $x \in A + B$ onto K . The first stage decomposes that projection into a common component in $A \cap B$ and a component polar to $A + B$. Moreau uniqueness then shows that the projection onto K is exactly the common component. Hence

$$\{P_K x : x \in A + B\} = A \cap B.$$

Restricting the source set to A or B gives the same image, and closed conic hulls yield the three cone equalities.

The formal proof follows this architecture rather than replacing it with finite-dimensional polyhedral computation. This is important: the theorem retains the general quantification stated in the source.

6.3 Algebraic consequences and circuits

The exact cone layer also verifies:

$$\begin{aligned} A \perp B &\subseteq B, \\ A \subseteq B &\iff B^\circ \subseteq A^\circ, \\ (A^\circ)^\circ &= A, \\ A \perp (A \perp B) &= A \perp B, \\ A \cap B &= (A^\circ + B^\circ)^\circ. \end{aligned}$$

Reflection commutes with rejection:

$$-(Y \setminus_{\perp} X) = (-Y) \setminus_{\perp} (-X).$$

These identities justify source-level circuits that implement projection with two rejection primitives and intersection with three rejection primitives, one standalone reflection, and closed cone sums.

7 Comparison and approximate invariance

Provenance. The comparison and deterministic invariance formulas come from the paper. Lean makes coverage and margin hypotheses explicit and proves the scalar and Gram inequalities; it does not add the omitted random-embedding law that would generate those hypotheses.

7.1 Comparing two cones

For a nonzero cone A , let

$$U(A) = \{x \in A : \|x\| = 1\}.$$

The directed similarity from A to B is

$$s_{\rightarrow}(A, B) = \inf_{x \in U(A)} \|P_B x\|,$$

and the symmetric similarity is

$$s(A, B) = \min\{s_{\rightarrow}(A, B), s_{\rightarrow}(B, A)\}.$$

The angle is $\theta(A, B) = \arccos s(A, B)$. The formalization proves $0 \leq s \leq 1$, identity on nonzero cones, and the right-angle consequence when one cone contains a direction polar to the other. The zero cone is handled separately, as in the source.

A normalized finite frame gives a computable minimum $\widehat{s}$. It is always optimistic:

$$s(A, B) \leq \widehat{s}(A, B).$$

If every unit direction in each cone lies within distance γ of a normalized frame ray, nonexpansiveness of metric projection gives

$$0 \leq \widehat{s}(A, B) - s(A, B) \leq \gamma.$$

The coverage premise is named and visible. No sampled biological frame is assumed to satisfy it automatically.

7.2 Scaled near-isometry

Let $M : H \rightarrow H'$ be a bounded linear map. On a relevant set S , a scaled near-isometry has parameters $\alpha > 0$ and $0 \leq \rho < 1$:

$$(1 - \rho)\alpha\|u\|^2 \leq \|Mu\|^2 \leq (1 + \rho)\alpha\|u\|^2, \quad u \in S.$$

When S is a real subspace, a balanced polarization argument proves

$$|\langle Mu, Mv \rangle - \alpha\langle u, v \rangle| \leq \rho\alpha\|u\| \|v\|.$$

This preserves a positive or negative inner-product sign whenever its unscaled magnitude exceeds $\rho\|u\|\|v\|$. Boundary points with zero margin are correctly excluded from a uniform sign guarantee.

7.3 Projection error

Let $p = P_B x$, and suppose $q \in B$ has transformed-space optimality

$$\|M(x - q)\| \leq \|M(x - p)\|.$$

If the near-isometry covers both residuals $x - q$ and $x - p$, then

$$\|x - q\|^2 \leq \frac{1 + \rho}{1 - \rho} \|x - p\|^2.$$

Metric-projection optimality gives the Pythagorean inequality

$$\|x - q\|^2 \geq \|x - p\|^2 + \|q - p\|^2.$$

Combining them yields the verified error theorem

$$\|q - p\| \leq \sqrt{\frac{2\rho}{1 - \rho}} \text{dist}(x, B).$$

The Lean endpoint takes feasibility, transformed optimality, and residual coverage as explicit premises. It does not package the source's preceding closed-image argument that constructs q as the preimage of $P_{MB}(Mx)$, nor the subsequent norm bound after reapplying M . The scalar calculation itself is fully checked.

7.4 Sparse Gram bounds

Let m_i be the columns of a transmission map, with common scale $\alpha > 0$. Suppose

$$\left| \frac{\|m_i\|^2}{\alpha} - 1 \right| \leq \eta, \quad \frac{|\langle m_i, m_j \rangle|}{\alpha} \leq \mu \quad (i \neq j).$$

For coefficients supported on a finite set I , define

$$M_I x = \sum_{i \in I} x_i m_i.$$

The formal proof expands the squared norm through the active Gram matrix. The diagonal error contributes at most $\eta\alpha \sum_i x_i^2$. For the off-diagonal part, the elementary inequality

$$2|x_i x_j| \leq x_i^2 + x_j^2$$

counts exactly $|I| - 1$ partners for each active coordinate. Consequently,

$$\left| \|M_I x\|^2 - \alpha \|x\|_2^2 \right| \leq (\eta + (|I| - 1)\mu)\alpha \|x\|_2^2.$$

For $|I| \leq k$, this gives the source's restricted-isometry bound with $\rho_k = \eta + (k - 1)\mu$. The proof is deterministic and uses no random matrix theorem; its role here is the deterministic geometric part of the sparse-recovery framework associated with restricted isometries [5].

Boundary of the invariance layer

The random Johnson–Lindenstrauss dimension estimate, exact equivariance of the complete cone algebra, the branch-incidence specialization $\mu \leq r/d$, and the heterogeneous spatiotemporal path model are not formal endpoints. The verified result begins with a bounded linear map and explicit deterministic geometric premises.

8 Signals, filters, and membrane dynamics

Provenance. The signal equations, wavelet parameters, and low-pass approximation are source material. Selecting restricted-measure Bochner L^2 , choosing causal truncation, constructing a compact Volterra operator, and replacing the $O(\Delta t^2)$ term by a Lipschitz-based bound are explicit formalization choices and strengthenings.

8.1 The recording-window signal space

Lebesgue measure is restricted to the closed interval $[0, T]$. Scalar and finite-vector signals are Bochner L^2 equivalence classes over that measure. A finite spatial matrix is lifted pointwise to a bounded linear map on the signal space, with an operator-norm bound inherited from the spatial map.

Finite nonnegative aggregation is verified in an arbitrary real normed signal space. If all component outputs lie in a closed cone and all weights are nonnegative, their finite weighted sum lies in the same cone. Temporal filters are represented as bounded linear operators, and preservation of a particular cone is a visible predicate rather than a consequence of boundedness alone.

8.2 A compact causal convolution model

The general source discussion uses finite-window convolution with suitable square-integrable kernels. Boundary conventions matter, so the development constructs one nontrivial model explicitly: the causal prefix-integration operator. The ambient Hilbert-space and compact-operator facts are standard functional analysis [4]; the point at issue is the exact finite-window realization.

$$(V_T f)(t) = \int_0^t f(s) ds \quad \text{for almost every } t \in [0, T].$$

For $T \geq 0$, the formalization proves

$$\|V_T\| \leq T.$$

It represents the operator by an L^2 kernel, approximates that kernel by simple functions, converts each simple kernel into a finite-rank operator, and obtains finite-rank approximation in operator norm. Compactness follows.

This model is identified with a causal unit-step impulse response under causal truncation. Separate theorems state that it is not the periodic or whole-line zero-extension convention. The generic Laplace transfer identity remains an interface; the concrete construction does not pretend to prove a general transform theorem.

8.3 An exact finite wavelet dictionary

The source's three wavelets are formalized in $L^2([0, T])$ with

$$T = 0.20, \quad h(t) = \begin{cases} \sin^2(\pi t/T), & 0 \leq t \leq T, \\ 0, & \text{otherwise,} \end{cases} \quad \gamma = \frac{4}{\sqrt{3T}}.$$

The atoms are

$$\begin{aligned} \phi_1(t) &= \gamma h(t) \cos(2\pi 30t), \\ \phi_2(t) &= \gamma h(t) \sin(2\pi 30t), \\ \phi_3(t) &= \gamma h(t) \cos(2\pi 60t). \end{aligned}$$

Exact Fourier integral calculations prove

$$\int_0^T \phi_i(t) dt = 0, \quad \langle \phi_i, \phi_j \rangle = \delta_{ij}.$$

The unconstrained, nonnegative, and plasticity examples then reduce to exact coordinate calculations. The verified residual and neuronal-output signs match the source. Literal firing-rate baselines are kept separate from the baseline-subtracted L^2 signals used in projection.

8.4 A quantified membrane update

Let $d(s)$ be the net excitation-minus-inhibition drive over a local step, $\tau > 0$, and $\Delta t \geq 0$. The exact first-order low-pass response is

$$z(\Delta t) = e^{-\Delta t/\tau} z_0 + \int_0^{\Delta t} \frac{1}{\tau} e^{-(\Delta t-s)/\tau} d(s) ds.$$

With

$$\lambda = 1 - e^{-\Delta t/\tau},$$

the integral of the kernel is exactly λ . Adding and subtracting the frozen drive $d(0)$ yields

$$z(\Delta t) = (1 - \lambda)z_0 + \lambda d(0) + r.$$

If d is L -Lipschitz on the step, the formalization constructs the remainder and proves

$$\|r\| \leq \frac{L}{\tau} |\Delta t|^2.$$

Thus the source's $O(\Delta t^2)$ is not an opaque annotation. It has an explicit witness and bound. The premise is deliberately not inferred for an arbitrary L^2 signal, because an L^2 equivalence class need not have pointwise Lipschitz regularity.

9 Nonnegative least squares and learning

The optimization layer uses the classical nonnegative least-squares problem [3], but verifies the particular fixed-point, optimality, and convergence claims needed by the model instead of importing an algorithm as a black box.

Provenance. The objective, clipped update, and learning interpretation come from the paper. The exact finite KKT equivalence and the two deterministic convergence packages are formalization contributions. No stochastic convergence theorem is attributed to either the source or the Lean development.

9.1 Finite objective, gradient, and projection

Let $X \in \mathbb{R}^{n \times N}$, $y \in \mathbb{R}^n$, and $w \in \mathbb{R}^n$. The finite batch objective is

$$F(w) = \frac{1}{2} \|X^\top w - y\|_2^2, \quad w \geq 0.$$

Its gradient is

$$g(w) = X(X^\top w - y).$$

The formalization proves the exact quadratic expansion

$$F(w + h) = F(w) + \langle g(w), h \rangle + \frac{1}{2} \|X^\top h\|_2^2,$$

and proves that $g(w)$ is the directional derivative in every coefficient direction.

Projection onto the nonnegative orthant is componentwise positive part. For $\varepsilon > 0$, define

$$T_\varepsilon(w) = P_{\mathbb{R}_+^n}(w - \varepsilon g(w)).$$

The formal theorem states

$$T_\varepsilon(w) = w \iff w \geq 0, \quad g(w) \geq 0, \quad w_i g_i(w) = 0 \text{ for every } i.$$

The conditions on the right are the complementary KKT conditions. Using the quadratic expansion, the development proves both directions of

$$\text{KKT point} \iff \text{global NNLS minimizer}.$$

9.2 Deterministic convergence

Orthant projection is nonexpansive. Under lower and upper spectral bounds

$$\mu \|h\|_2^2 \leq \|X^\top h\|_2^2, \quad \|Xz\|_2^2 \leq L \|z\|_2^2,$$

with $0 < \mu \leq L$ and $0 < \varepsilon < 2/L$, the unprojected error map has contraction factor

$$q = \sqrt{1 - \varepsilon(2 - \varepsilon L)\mu} < 1.$$

The projected iteration inherits this contraction and converges geometrically to the unique coefficient minimizer.

The formal result is stronger in the rank-deficient case than a simple contraction proof. With only an upper spectral bound, a nonempty solution set, $L > 0$, and $0 < \varepsilon < 2/L$, it proves:

- Fejer descent toward every minimizer;
- summability and vanishing of prediction error;
- convergence in finite coefficient space to some NNLS minimizer;
- asymptotic regularity of successive iterates; and
- equality of the predictions produced by all minimizers.

Coefficient uniqueness is recovered only when $w \mapsto X^\top w$ is injective. Kernel directions are not incorrectly collapsed.

9.3 The stochastic boundary

One application of the clipped online rule is exactly a projected stochastic-gradient step when the output is the instantaneous residual. Convergence of a stochastic sequence is a different theorem. It requires a specified probability space or filtration, unbiased sampling, moment bounds, and step-size conditions. The source states these as standard qualifications; the Lean project does not invent the missing stochastic model. General projected stochastic-gradient results illustrate the kind of additional hypotheses that such a theorem needs [6].

Similarly, a constant small learning rate generally tracks a solution with steady-state misadjustment rather than converging exactly. When the membrane state has $\lambda < 1$, the transient product is an exact gradient only for a correspondingly matched filtered objective. These limits are part of the formal interpretation, not footnotes to be discarded.

10 Finite collateral occupancy

Provenance. The paper supplies the occupancy and overlap formulas as model calculations. The explicit uniform finite experiment, edge-case theorems, exponential approximation error bound, and independent product construction are constructive strengthenings added by the formalization.

The occupancy calculation is formalized as a finite probability experiment, implemented by exact uniform averages over finite sets. One active axon chooses a p -element subset of n targets. A sample of m axons is a function assigning one such subset to each axon; the Cartesian product makes the choices independent.

For a fixed target, the miss probability for one axon is $1 - p/n$, hence the miss probability for all m axons is $(1 - p/n)^m$. Summing target indicators gives:

Theorem 10.1 (Exact occupied-target expectation). *For $0 < n$ and $p \leq n$, the expected number of occupied targets is*

$$q = n \left(1 - \left(1 - \frac{p}{n} \right)^m \right).$$

Zero targets, zero axons, zero collaterals, and invalid $p > n$ cases have separate theorems, so division and empty sample spaces are not hidden inside the main statement.

The source replaces the finite miss probability by an exponential. Define

$$q_{\text{exp}} = n \left(1 - e^{-mp/n} \right).$$

The formalization proves the signed and absolute bounds

$$q_{\text{exp}} \leq q, \quad |q - q_{\text{exp}}| \leq \frac{mp^2}{n}.$$

This replaces an informal approximation sign by a uniform finite inequality.

For two messages sampled independently from the same product experiment, targetwise indicator independence yields the exact overlap formula

$$\mathbb{E}|O_1 \cap O_2| = \frac{q^2}{n}.$$

The theorem is exact for that product law. Its biological use still depends on the modeling approximation that distinct messages have the required independence and uniform marginals.

11 Population-level realization

Provenance. Active faces, population maps, and primitive circuits are source concepts. The finite KKT/Moreau certificate is a sufficient formalization lemma, while approximation quality, support selection, and biological implementation remain named premises rather than new theorems about neural tissue.

11.1 Active faces

For a finite generator family g_i , let C_I be the closed cone generated by indices I , and let $J \subseteq I$ be a candidate active set. Orthogonal projection onto the span of $\{g_j : j \in J\}$ is a candidate piece of the nonlinear cone projection.

The formal certificate requires:

1. the span projection has a nonnegative expansion in the selected generators g_j ;
2. the residual pairs non-positively with every ambient generator $g_i, i \in I$.

These are finite feasibility and polar inequalities. They show that the candidate point lies in C_I , the residual lies in C_I° , and the two are orthogonal. Moreau uniqueness then proves

$$P_{C_I}x = P_{\text{span}\{g_j : j \in J\}}x.$$

This is a constructive sufficient form of the source's local-face equation. Non-strict inequalities deliberately admit boundary overlap. A separate theorem proves that any two valid certificates at the same point return the same cone projection.

The development does not prove that these regions are exactly the intrinsic relative interiors of all polyhedral faces, nor that a biological threshold mechanism selects them. Those are stronger geometric and empirical claims.

11.2 Approximate population maps

A population region records:

- a source cone and target cone;
- a modeled message family;
- one active region;
- the corresponding face-span projection; and
- a support-selected population map T .

The source's learned approximation premise is exposed as

$$\|T(u) - P_{\text{span}(F)}u\| \leq \epsilon\|u\|$$

on the relevant intersection of source cone, message family, and active region. Since active-face certification gives $P_{\text{span}(F)}u = P_Qu$, Lean proves

$$\|T(u) - P_Qu\| \leq \epsilon\|u\|$$

and the identical residual error

$$\|(u - T(u)) - (u - P_Qu)\| \leq \epsilon\|u\|.$$

In the exact zero-error case, if the complete source cone is covered, the closed conic hulls of outputs and residuals are exactly

$$\text{Conic}(T(P)) = P \perp Q, \quad \text{Conic}\{u - T(u) : u \in P\} = P \setminus_\perp Q.$$

What equation (204) means formally

The learned approximation bound is a premise. Lean proves its consequences and prevents it from being hidden inside a definition asserting that a neuron implements projection. It does not derive the numerical value of ϵ from anatomy, training data, or synaptic dynamics.

11.3 Primitive circuits and conditionals

The ideal output cone of one population is the reflected rejection

$$\mathcal{P}(P, Q) = -(P \setminus_\perp Q).$$

Using reflection equivariance, double rejection, and the intersection identity, the development verifies the exact projection and intersection circuits described in the source.

The conditional circuit has two explicit premises:

- the control signal is inactive exactly when the condition cone is the zero cone;
- sufficiently strong synchronized inhibition passes the signal when control is inactive and blocks it otherwise.

Under those premises, the output is

$$\begin{cases} -(P \sqcup Q), & C = \{0\}, \\ \{0\}, & C \neq \{0\}. \end{cases}$$

No cone-emptiness detector, synchronization mechanism, or inhibitory dynamics is smuggled into the proof.

12 Verification evidence and traceability

12.1 Authoritative-source audit

The distributed paper was not the sole transcription source. The final formal state was audited against the author-supplied archive `arXiv-2309.02332v3.tar.gz`, whose SHA-256 digest is

`a85f550647583374b73549e34cff596d0bd0c7991ea5586583f6cbdb068e1305.`

The archive contains one top-level TeX source, an embedded bibliography, and twelve PDF figures. Three independent `pdflatex` passes produced a 60-page document with no unresolved citation or cross-reference. A layout-preserving text diff against the supplied arXiv PDF has one hunk: the arXiv identifier and its induced title-page positioning. The article body is identical.

The audit reconstructed formula bodies, labels, macro meanings, and nearby qualifications through equation (210). It found no material mathematical discrepancy in the claims made by the Lean project. One module-header section reference was corrected, and the transmission-invariance limitations were made more explicit in the project documentation.

12.2 Build, gaps, and axioms

The complete build checks 2,960 build jobs. The project contains 31 Lean modules and 8,367 lines of Lean source. A direct scan finds no `sorry`, `custom axiom`, or `unsafe declaration` in project modules.

An explicit `#print axioms` audit covers 38 public endpoints across subspaces, cones, approximation, optimization, signals, wavelets, population realization, circuits, and probability. Every endpoint depends only on

`{propext, Classical.choice, Quot.sound}`.

The job count and line count are reproducibility facts, not measures of mathematical importance. The substantive evidence is the checked statement of each endpoint and its source traceability.

12.3 Coverage summary

Table 5: Scientific coverage of the formal development.

Layer	Status	Verified content and boundary
Finite subspaces	Verified	Column-space identities, Moore–Penrose construction and uniqueness, projection/rejection/intersection renderings, Frobenius and trace comparisons, and non-distributivity. Gleason’s theorem is cited background.
Closed cone core	Verified	Closed conic hull, non-positive polar, metric projection, Moreau decomposition, closed sum, projection, rejection, reflection, exact cone laws, double rejection, and intersection.
Cone comparison	Verified	Inclusion by zero rejection, exact worst-case similarity and angle, finite frame proxy, and the γ -coverage error theorem.
Approximate invariance	Conditional	Near-isometry, polarization, angular margin, scalar projection error, and sparse Gram/RIP bounds. Random embeddings, exact full equivariance, and the anatomical branch specialization are outside the endpoints.
Temporal signals	Conditional	Recording-window L^2 , spatial lifting, finite aggregation, temporal operator interfaces, compact causal unit-step Volterra filtering, exact wavelets, and the Lipschitz membrane remainder. Generic Laplace and kernel theory remain interfaces or outside scope.
NNLS and learning	Verified	Finite objective and gradient, projected fixed points, KKT equivalence, full-rank contraction, rank-deficient convergence, prediction uniqueness, and exact wavelet instances. General stochastic convergence is outside scope.
Collateral occupancy	Verified	Exact finite expectation, explicit exponential error bound, and exact two-message overlap under an independent product law. Biological independence remains a model assumption.
Population realization	Conditional	Constructive sufficient active-face regions, consequences of the learned approximation bound, exact hulls at zero error, primitive circuits, and the abstract conditional. Training, support selection, gating dynamics, and empirical error values are not derived.

13 Scientific interpretation and limitations

13.1 What has been established

The formalization establishes that the cone algebra is internally coherent under a precise topological convention and that its two longest new identities are valid in the generality stated by the source.

It also establishes that several surrounding quantitative claims can be made exact: the occupancy experiment has a closed expectation formula, its exponential replacement has an explicit finite error, the wavelet dictionary is exactly orthonormal, the membrane remainder has a quadratic bound under Lipschitz drive, and deterministic projected NNLS converges even when coefficient minimizers are not unique.

The result is more than a transcription check. Important hidden assumptions were either removed or exposed:

- pseudoinverse formulas cover rank-deficient matrices;
- conic images use closed hulls explicitly;
- nonzero hypotheses appear in similarity measures;
- residual coverage appears in approximate-invariance theorems;
- temporal boundary conventions are distinguished;
- Lipschitz regularity is not inferred from L^2 ;
- stochastic and deterministic convergence are separated; and
- population approximation and biological control remain premises.

13.2 What has not been established

The development does not show that a mammalian neural population represents a cognitive concept by a convex cone. It does not validate the mechanistic single-neuron model, estimate ρ , ϵ , γ , or Lipschitz constants from data, or prove recurrent-network stability. It does not formalize every equation in the source, every cited theorem, or every illustrative population example.

The formal claim is narrower and sharper:

If messages, maps, filters, learning problems, and population regions satisfy the formal interfaces stated here, then the verified cone-algebra, approximation, optimization, and occupancy conclusions follow.

This is the appropriate role of theorem proving in a mechanistic modeling program. It isolates logical validity from empirical adequacy, allowing future data or richer models to target named premises rather than an informal chain of reasoning.

13.3 Consequences for future work

The most valuable extensions are not simply more lines of formal proof. They are bridges that reduce scientifically important premises:

- a concrete transform theorem connecting biological impulse responses to the Laplace-domain interface;
- a probability model for stochastic learning with explicit filtration and moment assumptions;
- an intrinsic polyhedral-face theory connecting certificates to relative-interior partitions and threshold support selection;
- certified data pipelines that turn measured connectivity and responses into bounds on ρ , μ , η , and ϵ ; and
- network-level stability theorems for recurrent compositions of the verified population primitives.

Each extension would change the scientific boundary, not merely expand the formal library.

14 Conclusion

The cone-algebra model has a substantial deterministic core that is both mathematically meaningful and amenable to machine verification. The formal development confirms Moreau-based projection and rejection, the double rejection identity, the rejection-only intersection formulas, and a broad supporting theory spanning subspaces, approximation, signals, optimization, finite probability, and conditional population realization.

The strongest conclusion is therefore conditional but rigorous. Within the explicit model class, neural population messages can be represented and manipulated by an algebra of closed convex cones, and the central algebraic claims are logically correct in arbitrary real Hilbert spaces. Lean checks that conclusion without admitted proofs or project-specific axioms. Whether the model describes biological computation with useful fidelity remains a question for empirical science and for future formal bridges from data to the named assumptions.

A Source equations and principal Lean endpoints

The table below is a compact index. Internal helper lemmas are omitted. Full qualification and negative coverage are recorded in the source-conformance audit.

Table 6: Selected source-to-Lean traceability.

Source	Mathematical content	Principal endpoint or artifact
(4)–(11)	Column spaces and $\text{Col}(A) = \text{Col}(AA^T)$	matrixColumnSpace_mul_transpose
(12)–(23)	Subspace complement, projection, rejection, and intersection	double_subspaceRejection_eq_projection; intersection_eq_sum_reject_mutualRejections
(24), (27)–(29)	Frobenius, trace, directional overlap, and cosine	subspaceOverlap_eq_frobeniusInner; symmetricSubspaceCosine_eq_frobenius
(30)	Occupancy expectation, exponential replacement, and independent overlap	expectedOccupied_eq_exactFormula; expectedOccupied_exponential_sq_error; expectedOccupiedOverlap_eq_sq_div
(31)	Nonnegative finite signal aggregation	aggregateSignal_mem_cone
(33)	Low-pass step and quadratic remainder	exactMembraneUpdate_eq_updateWithRemainder
(36), (39)–(41)	Finite NNLS, gradient, fixed point, KKT, and global optimality	nnlsLoss_hasDerivAt_line; nnlsMinimizer_iff_complementaryKKT
After (41)	Full-rank and rank-deficient deterministic convergence	projectedBatchIterate_tendsto_of_designSpectralBounds; projectedBatchIterate_tendsto_some_nnlsMinimizer_of_upperBound_of_exists
(62), (65)–(71)	Closed conic hull, metric projection, polar, and Moreau decomposition	conicHull; metricProjection_eq_iff; moreau_unique
(72)–(80)	Closed sum, conic projection, rejection, intersection, and reflection	coneSum; coneProjection; coneRejection; coneReflection
(87)–(92)	Exact and finite-frame cone comparison	frameConeSimilarity_error92
(93)–(97)	Exact cone laws	coneProjection_absorb; inf_eq_polar_coneSum_polar
(98)–(108)	Projection by double rejection	metricProjection_polar_rejection_eq; doubleRejection
(109)–(157)	Intersection by rejections and sums	intersectionByRejections
(161)–(163)	Scaled near-isometry, polarization, and sign margin	scaledInnerDistortionOn_of_scaledNearIsometryOn_submodule; inner_map_pos_of_margin; inner_map_neg_of_margin
(165)–(167)	Projection-error calculation	projection_error_of_scaledNearIsometryOn
(170)–(171)	Gain/coherence and sparse Gram bound	sparseGramRIP171_of_card_le
(176)	Ideal population primitive	populationPrimitive
(195)–(196)	Span reduction and nonnegative homogeneity	metricProjection_starProjection_eq; metricProjection_smul

Source	Mathematical content	Principal endpoint or artifact
(199)	Constructive active-face projection	<code>metricProjection_finiteGeneratedCone_eq_selectedFaceProjection</code>
(204), (206)–(208)	Population approximation premise and verified consequences	<code>PopulationRegion.Approximation204;</code> <code>PopulationRegion.projection_error206;</code> <code>exact_population_hulls</code>
(209)–(210)	Reflection/rejection and conditional circuit	<code>coneReflection_rejection;</code> <code>controlledConditional210</code>

B Notation and assumption checklist

Symbol or phrase	Meaning in this report
H	An arbitrary real Hilbert space unless a finite Euclidean or recording-window space is stated.
C°	The non-positive polar $\{z : \langle z, c \rangle \leq 0 \ \forall c \in C\}$; the source prints $\neg C$.
$\text{Conic}(S)$	The norm-closed conic hull, not merely finite conic combinations.
$P_C x$	The unique metric projection onto a closed convex cone; a noncomputable mathematical choice, not extracted numerical code.
$A \perp B$	Closed conic hull of pointwise projections of A onto B .
$A \setminus_\perp B$	Closed conic hull of residuals $a - P_B a$; the paper’s binary rejection.
Finite cone	Finitely generated cone. The headline theorems need only closed convexity.
Near-isometry	A deterministic two-sided squared-norm bound on every relevant intermediate vector, not only on observed frame vectors.
Verified population map	A theorem conditional on a support-selected map and approximation premise, not an empirical claim about a recorded population.

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